

Specialized LLM Bot Using Pre-Trained Models

Project Summary

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Please paste the GitHub Repo link.
GitHub Profile Link: https://github.com/AniketGhorpade
GitHub Repository Link: https://github.com/AniketGhorpade/Specialised_LLM_Finance_Chatbot_Using_PreTrained_Models

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Summary

Overview

- In this capstone project, we will embark on an exciting adventure to construct an Industry-Specific Large Language Model (LLM) Bot using state-of-the-art pre-trained models from sources like Hugging Face. The primary objective is to develop an intelligent bot that can efficiently engage with people by answering their questions and providing information relevant to their sector. This project will help you become more technically proficient while also giving you a comprehensive understanding of the nuances, challenges, and trends of the chosen business.

- **Project Goals:**

- **Industry Selection:** Pick one sector to concentrate on for the duration of the project, which will involve gathering data and training the model.

- **Data Collection:** Gather relevant data specific to the chosen industry. This data will be used to improve the pre-trained model and ensure that the LLM Bot is contextually aware and knowledgeable about industry-specific information.

- **Model Selection and Training:** Utilize pre-trained models from any Hugging Face or similar platform. Adjust the model based on the collected data using programs like Google Colab with T4 GPUs. Keep the training time to a maximum of 25 epochs to guarantee viability.

- **Bot Development:** Build an LLM bot that can converse with users, responding to their inquiries and engaging in thought-provoking conversations pertinent to the chosen industry. The bot should demonstrate its ability to understand and effectively respond to industry-related queries.

- **Demonstration:** Create an instructional movie that describes the LLM Bot's workings. The video should highlight the bot's ability to answer questions specific to the company in order to illustrate its value.

Problem Statement:

- This capstone project's goal is to use pretrained models from websites like Hugging Face to develop an industry-specific Large Language Model (LLM) Bot. Gather information and refine an LLM that has already been trained.
- Provide accurate and contextually appropriate responses to demonstrate the bot's ability to communicate with users effectively.

Abstract

This project shows the entire process of creating a finance chatbot tailored to a particular industry using Hugging Face's pre-trained models.

The chatbot is designed for accurate and efficient responses by using sophisticated approaches like PEFT, LoRA, and QLoRA, as well as by finetuning the model using data related to finance. A user

friendly interface is guaranteed by the chatbot's deployment utilizing Flask and Streamlit, and its useful application in the finance sector is demonstrated in detail.

The chatbot's capabilities could be further expanded in the future, making it a priceless resource for questions and insights unique to a given sector Industry Selection:

Chosen Industry:- Finance Reasons:

Demand for AI in Finance Is High:

Process Automation: AI is increasingly being used in the financial industry to automate routine tasks including transaction processing, fraud detection, and customer service.

Rich and Diverse Information Sources:

Various types of data: The finance sector uses a range of data types, including transactional data, market data, news articles, and sentiment from social media.

Complex and Specialized Language:

Industry-Specific Terminology: The financial industry has a unique language and complex concepts that require specialized management and expertise.

Enhancing the Customer Experience:

Tailored Investment Suggestions and Financial Advice: An LLM Bot can provide tailored financial advice and investment recommendations.

Dealing with Fraud and Security:

Fraud Detection: AI and LLMs are useful tools for spotting possible fraudulent activity and questionable trends.

The banking sector is a perfect fit for creating a custom LLM Bot because of its dynamic nature and crucial requirement for accuracy. With an emphasis on finance, this initiative seeks to apply AI to solve challenging problems, boost output, and improve user experiences in a crucial industry.

Gathering Datasets

To optimize the pre-trained model and make sure it is contextually aware and informed about financial facts, our Finance Industry-specific LLM Bot development process necessitates data collecting.

Data Sources and Collection Methodologies Datasets that are publicly available:

Face Hugging Datasets: To ensure high-quality, industry-specific data, Hugging Face's pre-existing datasets were employed.

Finance Alpaca 1K Test: A dataset containing finance-related prompts and responses.

- **Content:** Includes 1,000 examples of finance-related questions and answers.
- **Purpose:** Provides a solid foundation for understanding finance-specific dialogues and improving response accuracy
- **Alpaca Finance EN:** Another comprehensive dataset focused on financial dialogues and scenarios.
- **Content:** A comprehensive dataset containing various financial scenarios, user queries, and expert responses.
- **Purpose:** Enhances the model's ability to engage in meaningful financial conversations and provide contextually relevant insights.

Project Workflow for Building the Finance Chatbot

- **Project Planning and Setup**
 - **Define Objectives:** Clearly define the objectives of the project, including the desired capabilities of the chatbot.
 - **Select Industry:** Choose the finance industry as the focus for your chatbot.

- **Set Up Environment:** Install the relevant software and libraries, such as Hugging Face Transformers, Streamlit, and Python, and set up the required development environment.

- **Information Gathering**

- **Locate Data Sources:** Find the sources of data unique to finance. Hugging Face datasets, financial news articles, company reports, and market analysis studies are a few examples of this.

- **Compile datasets:** Compile the chosen datasets. For instance, you may use Hugging Face datasets such as ssbuild/alpaca_finance_en and poornima9348/finance-alpaca-1k-test.

- **Preprocessing of Data:** To make sure the data is structured and appropriate for training, clean and preprocess it. This covers formatting text, dealing with missing values, and eliminating superfluous information.

Choosing a Model

- **Pick a Pre-trained Model:** From Hugging Face, pick a pre-trained language model that can be adjusted for use with financial data. Models based on GPT-3, BERT, or comparable architectures are popular options.

- **Configure the Training Environment:** To guarantee effective processing, train models using platforms such as Google Colab with T4 GPUs.

- **Fine-Tuning the Model**

Introduction to Fine-Tuning Techniques:

- **PEFT (Parameter-Efficient Fine-Tuning):** Optimize the model by adjusting only a subset of parameters, making the process more efficient.
- **LoRA (Low-Rank Adaptation):** Decompose the model parameters into lower-rank matrices for efficient fine-tuning.
- **QLoRA (Quantized LoRA):** Combine quantization with LoRA to further reduce memory usage and computation.
- Fine-tune the selected pre-trained model on the finance datasets using PEFT, LoRA, and QLoRA techniques.
- Limit training to a maximum of 25 epochs to ensure feasibility and manage computational resources.

- **Pushing Fine-Tuned Model to Hugging Face:**

- After fine-tuning, push the updated model to Hugging Face Model Hub for easier access and deployment.

- **Chatbot Development**

Construct the Flask Backend:

- Develop a Flask application to manage user inquiries and serve the optimized model.
- To test and demonstrate the Flask application, use ngrok to make it available online.

Construct the Streamlit Frontend:

- To give a user-friendly interface for communicating with the chatbot, develop a Streamlit application.
- Verify that the Streamlit app can use the ngrok URL to connect to the Flask backend.

• Design of User Interfaces

- **Develop User Interface:** Using Streamlit, create an intuitive user interface that enables people to communicate with the chatbot. Features like text input, response display, and discussion history ought to be part of this.

- **Include Features:** Include extra features such a button to delete chat history, a sidebar for entering the ngrok URL, and prompts to direct user engagement.

• Implementation

- **Host Backend:** The Flask backend is hosted via ngrok and made available via a public URL.

- **Frontend Deployment:** Make sure the hosted backend and the Streamlit frontend are able to communicate with each other.

Conclusion

This project shows the entire process of creating a finance chatbot tailored to a particular industry using Hugging Face's pre-trained models. The chatbot is designed for accurate and efficient responses by using sophisticated approaches like PEFT, LoRA, and QLoRA, as well as by fine-tuning the model using data related to finance. A user-friendly interface is guaranteed by the chatbot's deployment utilizing Flask and Streamlit, and its useful application in the finance sector is demonstrated in detail.

This finance chatbot's successful creation and implementation highlight how crucial it is to use cutting-edge AI approaches for effective fine-tuning. These methods not only improve the model's performance but also guarantee that it can accurately and contextually respond to queries unique to a certain industry.

The chatbot's capabilities could be further expanded in the future, making it a priceless resource for questions and insights unique to a given sector. The chatbot can stay up to date with the most recent facts and trends by continuously improving the model and growing its dataset, guaranteeing that it will continue to be a trustworthy and useful tool in the banking sector. This study highlights the potential of AI-driven conversational agents to revolutionize industry-specific communication and support, laying a solid platform for their future development.

Future Extent

- Adding more data sources to the chatbot to improve accuracy.
- Increasing the bot's capacity to handle increasingly difficult financial tasks.
- Modifying the chatbot for use in different sectors to show off its adaptability and scalability.